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SOLVED NUMERICALS – JUNE/JULY 2023 | 22S) (CSE STREAM)

1(C) The ratio of population of two energy levels is 1.059×10^{-30}. Find the wavelength of light emitted by spontaneous emission at 330K.

(CSE Stream, JUNE/JULY – 2023 | 22S) (Jan.2021 | July.2022 -18S)

Data: *The ratio of population,* $(N_2/N_1) = 1.059 \times 10^{-30}$

Ambient temperature, $T = 330K$

To find : *Wavelength of light emitted,,* $\lambda = ?$

Solution: we have Boltzmann factor, $\frac{N_2}{N_1} = e^{\frac{-\Delta E}{kT}}$

By Taking natural log on both sides, we have

$$In\left(\frac{N_2}{N_1}\right) = \frac{-\Delta E}{kT} = \frac{-hc}{\lambda kT} = -\left(\frac{hc}{k}\right)\left(\frac{1}{\lambda T}\right)$$

$$since\ \left(\frac{hc}{k}\right) = \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{1.38 \times 10^{-23}} = 0.014413$$

$$\lambda = -\left(\frac{hc}{k}\right)\left(\frac{1}{In\left(\frac{N_2}{N_1}\right)T}\right) = \frac{0.014413}{In(1.059 \times 10^{-30}) \times 330} = 632 \times 10^{-9} m = 632nm$$

∴ The wavelength of light emitted by spontaneous emission is 632nm.

2(C) The acceptance angle of an optical fiber is 30° when kept in an air find the angle of acceptance when it is in a medium of refractive index 1.33.

(Dec 2017 | 17S) (CSE Stream, JAN/FEB – 2023 | 22S) (CSE Stream, JUNE/JULY – 2023 | 22S)

Data: $Refractive\ index\ of\ core\ , n_0' = 1.33$

$Angle\ of\ acceptance\ , \theta = 30°$

To find: Angle of acceptance while the fiber is in the refracting medium, $\theta_0' = 1.33$

Solution: we have the equation, $sin\theta_0 = \frac{\sqrt{(n_1^2 - n_2^2)}}{n_0'}$ ……(1)

When the surrounding medium is air, then $n_0 = 1\ and\ \theta_0 = 30°$

$$sin30° = \sqrt{(n_1^2 - n_2^2)} \qquad 0.5 = \sqrt{(n_1^2 - n_2^2)} \text{.....(2)}$$

When the surrounding medium is of the refractive index 1.33, let θ_0' be the acceptance angle.

Now eqn(1) becomes, $\sin\theta_0' = \frac{\sqrt{(n_1^2 - n_2^2)}}{n_0'} = \frac{\sqrt{(n_1^2 - n_2^2)}}{1.33}$

$$1.33 \sin\theta_0' = \sqrt{(n_1^2 - n_2^2)} \text{ .....(3)}$$

$\therefore from\ equation(2) and\ (3),$ $1.33 \sin\theta_0' = 0.5$

$$\theta_0' = sin^{-1}\left(\frac{0.5}{1.33}\right) = 22°$$

∴ The Acceptance angle when the fiber is in the medium of refractive index 1.33 is 22°

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3(C) A particle of mass 0.5 MeV/c^2 has kinetic energy 100eV. Find its de-Broglie wavelength, where 'c' is the velocity of light.

(CSE Stream, JUNE/JULY – 2023 | 22S) [Dec. 2014 | Jan 2018, Dec 2019 | Feb 22 – 18S].

Data: Mass of the particle m = 0.5 MeV/c^2,

Kinetic energy E = 100 x 1.609 x 10^{-19} J

Velocity of light, $c = 3 \times 10^8\, m/s$

Velocity of neutron $v = \frac{1}{10} \times c = 3 \times 10^7 m/s$

To find: De Broglie wavelength, $\lambda = ?$

Solution: De Broglie wavelength, $\lambda = \frac{h}{mv}$

We have, mass of the particle, m = $0.5 \frac{\text{MeV}}{c^2} = \frac{0.5\times10^6\times1.602\times10^{-19}}{3\times10^8} = 8.9 \times 10^{-31}\ kg$

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 8.9 \times 10^{-31} \times 100 \times 1.602 \times 10^{-19}}} = 1.24 \times 10^{-10} m$$

$$\therefore \textbf{\textit{The De Broglie wavelength of the particle is}}\ \mathbf{1.24 \times 10^{-10} m}$$

4(C) In a measurement of position and velocity of an electron moving with a speed of 6x10^5m/s, Calculate the highest accuracy with its position could be determined , if the inherent error in the measurement of its velocity is 0.01% for the speed. **(CSE Stream, JUNE/JULY – 2023 | 22S)**

Solution: $\Delta v = \mathbf{6 \times 10^5} \times \frac{0.01}{100} = \boldsymbol{60\ m/s}$

$$\Delta x \geq \frac{h}{4\pi m \Delta v} = \frac{6.625 \times 10^{-34}}{4 \times 3.14 \times 9.11 \times 10^{-31} \times 60} = 9.65 \times 10^{-7}\ m$$

5(C). Given A=$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ Prove that $A^+ = A$

A=$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ $\qquad A^+ = (A^*)^T$

$A^* = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$ *and the* $(A^*)^T = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = A$, *hence proved* $A^+ = A$

7 (C). In a diffraction grating experiment the laser light undergoes second order diffraction, if the distance between screen and grating is 20cm and the average distance of the 2nd order spot is 2.7cm, calculate the wavelength of the laser light. Grating constant = 1×10^{-5}m.

(CSE Stream, JUNE/JULY - 2023 | 22S)

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Given: $m = 2$, $d = 1\times10^{-5}$ m, $f = 20$cm & $x =$ **2.7cm**

Angle of diffraction is given by $\tan\theta = \frac{x}{f}$

$$\therefore \theta = \tan^{-1}\left(\frac{x}{f}\right) = \tan^{-1}\left(\frac{2.7}{20}\right) = \tan^{-1}(0.135) = 7.69^{o}$$

Wavelength $\lambda = \frac{d\sin\theta}{m}$

$$\therefore \lambda = \frac{1\times10^{-5}\times\sin(7.69^{o})}{2}$$

$$\lambda = \frac{0.133\times10^{-5}}{2} = 0.0669\times10^{-5} = 669\times10^{-9}\,m \quad \text{i.e.,} \quad \lambda = \mathbf{669nm}$$

8(C) Calculate the angle of acceptance and numerical aperture of given optical fiber having diameter of the spot 2.6 cm and the distance between screen and optical fiber 3 cm.

(CSE Stream, JUNE/JULY – 2023 | 22S)

Given: $D = 2.6$ cm & $l = 3$ cm.

Angle of acceptance is given by $\tan\theta_{o} = \frac{D}{2l}$

$$\therefore \theta_{o} = \tan^{-1}\left(\frac{D}{2l}\right) = \tan^{-1}\left(\frac{2.6}{2\times3}\right) = \tan^{-1}(0.433) = \mathbf{23.4^{o}}$$

Numerical aperture NA = $\sin\theta_{o}$

NA = $\sin(23.4^{o}) = \mathbf{0.40}$

9(C) The number of particles emitted per second by a random radioactive source has a poison's distribution with $\lambda = 4$. Calculate the probability of $P(X = 0) and P(X = 1)$

$$P(X = 0) = \frac{\lambda^{k}e^{-\lambda}}{k!}$$

(K=X=0 and $\lambda = 4$)

$$P(X = 0) = \frac{4^{0}e^{-4}}{0\,!} = 0.0183$$

$P(X = 1)$,

$$P(X = 1) = \frac{4^{1}e^{-4}}{1\,!} = 0.0732$$

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10(C). A slowing in object in an animation has a first frame distance 0.5m and first slow in frame 0.35m. Calculate the base distance and the number of frames in sequence.

(CSE Stream, JAN/FEB – 2023 | 22S) (CSE Stream, JUNE/JULY – 2023 | 22S)

$$base\ distance = \frac{distance\ between\ any\ two\ adjacent\ frames}{2}$$

$$base\ distance = \frac{0.5 - 0.35}{2} = 0.075m$$

To find number of frames

$$no\ of\ frames = \frac{total\ distance}{base\ distance}$$

$$No\ of\ frames = \frac{0.5}{0.075} = 6.66 \approx 7\ frames$$

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